v = 500 km h^{-1} = 138.9 m s^{-1}

“horizontal circle” vertical equilibrium
$L \sin 50° = mg$ (1)

Centripetal force,

$$F_c = L\ cos\ 50° = \frac{mv^2}{r} \quad (2)$$

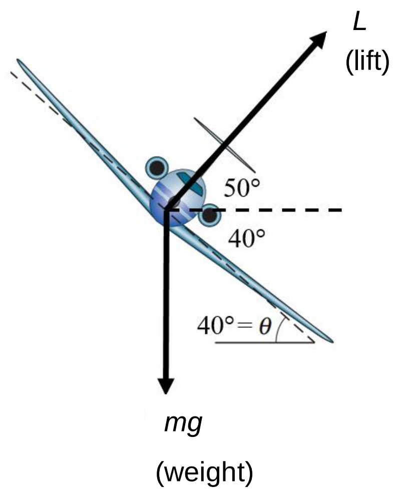
(1) / (2) $\dfrac{L \sin 50°}{L \cos 50°} = mg \div \dfrac{mv^2}{r}$

$$\tan 50° = \frac{gr}{v^2}$$

$$r = \frac{v^2}{g}\tan 50° = \frac{(138.9)^2}{9.81}\tan 50°$$

r = 2.3 km

Answer: D



9 Gravitational potential at infinity is zero